k-Permutation

- **Definition**
  A ***k*-permutation** (or ***k*-element permutation**) of a set of n distinct elements is an ordered arrangement of $k \leq n$ elements of the set.

- **Example**
  Let $S = \{1, 2, 3\}$. The ordered arrangement $(3, 2)$ is a 2-permutation of S.
- $(2, 3)$ is a different 2-permutation.
- **Definition**
  A ***n*-permutation** of a set of n distinct elements is simply called a permutation of the set.